DRUMPANEL · PROBLEM STATEMENT

DrumPanel problem statement

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-24	Amish Chadha	Kit 1.8 upgrade; users-and-needs and stakeholders tables added (Tables 1 and 2), drawn from the users, context and co-design notes; the user requirements DMP-URS-001 are drawn from them
0.3	2026-10-03	Amish Chadha	TRL 3 update; budget worded as a value-engineering target; open questions answered by the decisions of 2026-10-03 (DMP-DDR-001, DMP-DDR-002)
0.2	2026-10-03	Amish Chadha	TRL 2 update; drum facts, constraints and acceptance rule for drum contents added; first co-design candidate named
0.1	2026-09-30	Amish Chadha	Initial scaffold

DrumPanel problem statement

The first step in drum-based metal art, opening a drum into flat sheet, is still done by fire, chisel and body weight.

The problem

Accounts from Croix-des-Bouquets describe the same sequence: ends removed with chisel and hammer, the drum filled with dried leaves or cane and set on fire, the side split, and the body flattened by people climbing inside and pounding it down (SFO Museum; Roots of Development). It is apprentices who usually do this stage (Global Exchange). The fire is there to remove paint and residue; DrumPanel treats the smoke from burning unknown residues, and the noise of chisel work, as the things to design out.

In the preliminary IP screen the closest alternatives were powered drum cutter-flatteners and an expired US patent for a drum-head cutter (US3161952A). The screen found no simple, open, hand-powered bench set that combines a purge, a vapour check, cold cutting of the ends and seam, and rolling flat. The gap is a documented manual method that a workshop can build and trust.

Users and context

User	Need	Context
Metal artists and their apprentices	Flat, clean sheet from drums with less smoke, noise and effort	Small family workshops such as those in Croix-des-Bouquets
Local fabricators	Drawings for a bench set they can build from commodity steel	Towns with a welder and a lathe but no sheet metal machinery
Artisan cooperatives and fair trade buyers	A safer, repeatable preparation step they can fund and document	Export supply chains
Drum collectors and recyclers	A way to open drums that cannot be reused whole without burning them	Port and industrial areas

Users and needs

Table 1. Users and needs.

User type	Goals and needs	Typical use scenario	Pain points and challenges
Apprentices in drum art workshops	Open drums safely and quickly; flat sheet without climbing inside drums	Two apprentices prepare a morning's drums for the artists of a workshop cluster	Smoke from burning residue and paint; loud chisel work; tiring body-weight flattening; cut hands
Metal artists	Clean, flat sheet of a known size, and the head discs, ready to mark out	An artist collects three panels and two discs per drum for a sculpture or wall piece	Uneven, buckled sheet that needs hours of hammering; burnt paint and soot on the steel
Local fabricators	Drawings, a build plan and a bill of commodity parts they can build from	A town welding shop with a lathe builds one set for a cooperative and maintains it	No sheet metal machinery; imported tools cannot be repaired locally
Drum collectors and recyclers	A way to turn drums that cannot be reused whole into sheet, without burning	Drums from the port are sorted by their labels and the oil drums go to the bench set	Unknown contents; drums with fuel or chemical residue must be kept out

Stakeholders

Table 2. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
Artisan cooperatives and community arts organisations	Host and own a shared bench set; shape the procedure around how apprentices work	Co-design partner: trials, procedure review in the apprentices' language, panel sizes
Fair trade buyers and export partners (for example Singing Rooster)	A safer, documented preparation step in their supply chain; may fund a set	First contact to reach a cooperative; nothing agreed (DMP-DDR-001, D11)
Vocational welding schools	Build the first set as a training project; teach the vapour check	Second co-design candidate, for the build
Neighbours of the workshops	Less smoke and noise in the street	Informed through the cooperative
Local authorities (environment, labour)	Disposal of purge water and oil; workplace safety	To assess where the first set is placed (DMP-REG-001)
Gas detector suppliers	Supply, calibration gas and service of the LEL detector	Local supplier to confirm before purchase (TRL 4)

Operating environment

- Open-sided workshops or yards, hot and humid, often with no reliable mains power.
- Drums of 55 US gallon (about 208 L) steel, previously holding oils or unknown contents. A tight-head drum is about 572 mm inside diameter and 880 mm long over its rolled end seams (chimes), with two rolling hoops pressed out of the body and two screw bungs (50 mm and 20 mm) in one head. The body wall is usually 0.8 to 1.2 mm and the heads 1.0 to 1.5 mm.
- Workers standing at a bench; dust, rain and salt air near the coast.
- Water from a tap, a tank or a hose is available most days; electricity is not assumed.

Constraints

- Value-engineering target: USD 3,000 in parts for the full bench set (a hypothetical control target that keeps the design lean, not a spending limit).
- Manual operation only: hand cranks, levers and clamps, no motors.
- No flame, no grinding and no power tools on a drum until it has been purged and has passed the vapour check.
- Only drums whose label shows they held a non-volatile oil (lubricating, hydraulic or vegetable oil, flash point above 60 °C) are accepted. Drums that held fuel, solvent, chemicals or anything unknown, or that have no readable label, are rejected and returned to the supplier.
- Buildable by a local fabricator from commodity steel, bearings and fasteners; hardware under CERN-OHL-S-2.0.
- Purge and vapour check must be simple enough to follow from a printed pictogram sheet.
- Published as an open engineering reference, not as certified equipment.

Out of scope

- Cleaning drums for reuse as containers.
- Drums that held toxic, corrosive or unknown hazardous contents; these are not processed.
- Paint stripping and surface finishing beyond what the purge step removes (to be a separate question).
- Artistic cutting and embossing tools.

Prior work

Prior work	What it does	Gap for these users	Source
Croix-des-Bouquets burn and chisel method	Chisel off lids, split the side, burn out with dried cane or grass, flatten	Uses open fire and hand chisels	link
Banana-leaf burn-out with hammer and chisel	Drum stuffed with banana leaves and burned, then sliced and stretched to a 3 by 6 ft sheet using only hammers and chisels	Same fire step; no purge or vapour check	link
Body-weight flattening	Artists climb inside the split drum and use their weight to open it, then pound and sand it flat	Tiring and uneven; no rolling or clamping tool	link

Co-design

An artisan cooperative or community arts organisation working with drum-based metal artists, paired with a vocational training centre or local fabricator, to test the bench set in a working studio and shape the procedure around how apprentices actually work.

The first candidate to approach is a fair trade organisation that already works with Haitian metal artists, such as Singing Rooster, to reach a cooperative of Croix-des-Bouquets artists wherever they now work; a vocational welding school is the second candidate, for the build. Neither has been approached and nothing is agreed (DMP-DDR-001, D11).

Co-design checklist, to complete with the partner before any trial:

- [ ] Partner and studio agreed, with a named contact.
- [ ] Drum supply described: where drums come from, what they held, how they are labelled.

- [ ] Water supply and disposal route for purge water agreed.
- [ ] Pictogram procedure sheets reviewed with apprentices in their language.
- [ ] Panel sizes (about 233 x 1,800 mm) and head discs accepted by the artists.
- [ ] A person trained in the gas detector named for each session.

Open questions, answered on 2026-10-03

Each question below was answered by a decision made under Amish's pre-approval of 2026-10-03; the decision records give the reasons.

Question	Answer	Record
Burning does two jobs today, clearing residue and stripping paint. What replaces the paint stripping step without fire?	The panels leave the bench set painted. Paint removal is a separate cold step (scraper and wire brush, or a non-flammable stripper outdoors) outside this design; the bench set never heats the steel.	DMP-DDR-001, D4
Which purge method (water fill, air flush or steam) suits workshops with little water or power?	Water fill with a little detergent, soak 10 minutes, drain through the bottom bung; the water settles in a spare drum and is reused. Air flush needs a blower and steam needs fuel, so neither is used.	DMP-DDR-001, D1
What wall thickness range of drum steel must the cutter and rolls handle?	Body 0.8 to 1.5 mm and heads 1.0 to 1.5 mm; forces are sized at 1.5 mm and the slip roll setting is worked out for 0.8 to 1.2 mm.	DMP-DDR-001, D8; DMP-CAL-001
Would a shared bench set per cooperative work better than one per workshop?	One bench set shared by a cooperative or a cluster of workshops.	DMP-DDR-001, D9
Which artisan cooperative would host the first trials, given the security situation in Croix-des-Bouquets?	First candidate to approach: Haitian metal artists reached through a fair trade partner, wherever they now work; nothing agreed.	DMP-DDR-001, D11

Safety

> **Safety:** An empty drum that held oil can still hold flammable vapour, and cutting, grinding or heating it can make it explode. DrumPanel removes the fire from the process, but the vapour hazard stays until each drum is purged and checked. Drums that held fuel, solvent, chemicals or unknown contents are never processed. Cut drum steel has sharp edges, and the hand-cranked tools have in-running nips; the design answers these with guards, gloves and the safety stops in DMP-BLD-001. DrumPanel is published as an open engineering reference, never as certified equipment.